Introduction to Mobile Robotics

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What is a robot ?

Robot = *mechatronic system with perception, decision and action skills,* capable of carrying out *different tasks in the real world, in an autonomous way.*

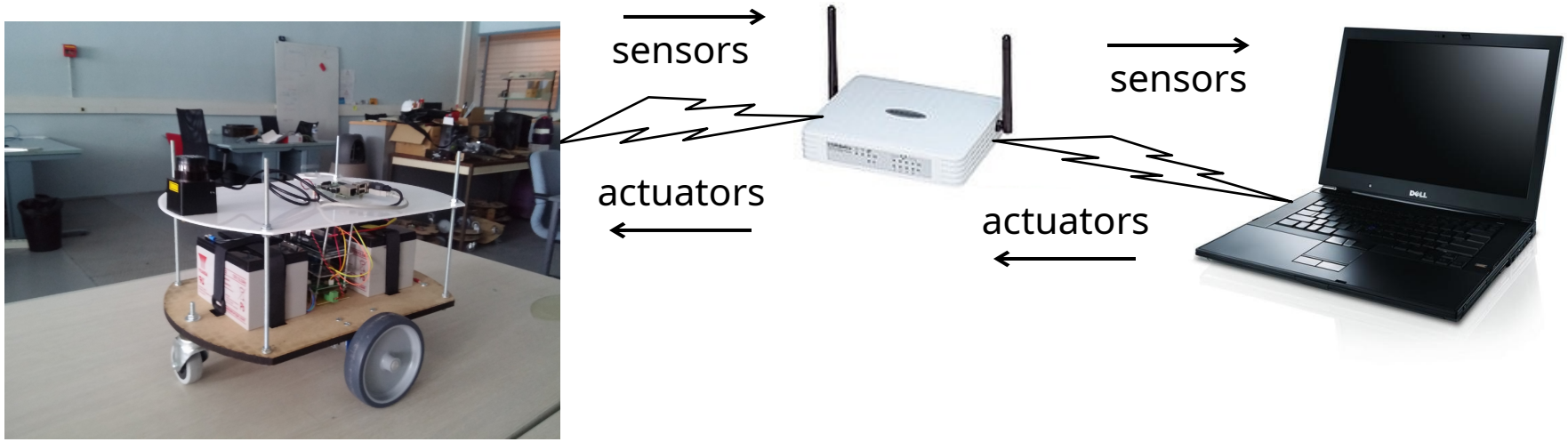






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The robot of the day



- 1 raspberry pi3:
  - ubuntu + ROS
- Sensors
  - 1 laserscanner
- Actuators
  - 2 wheels driven by 2 motors + encoders

- 1 PC Ubuntu + ROS
  - In charge of sensor data acquisition, processing & visualization;
  - In charge of controlling actuators.

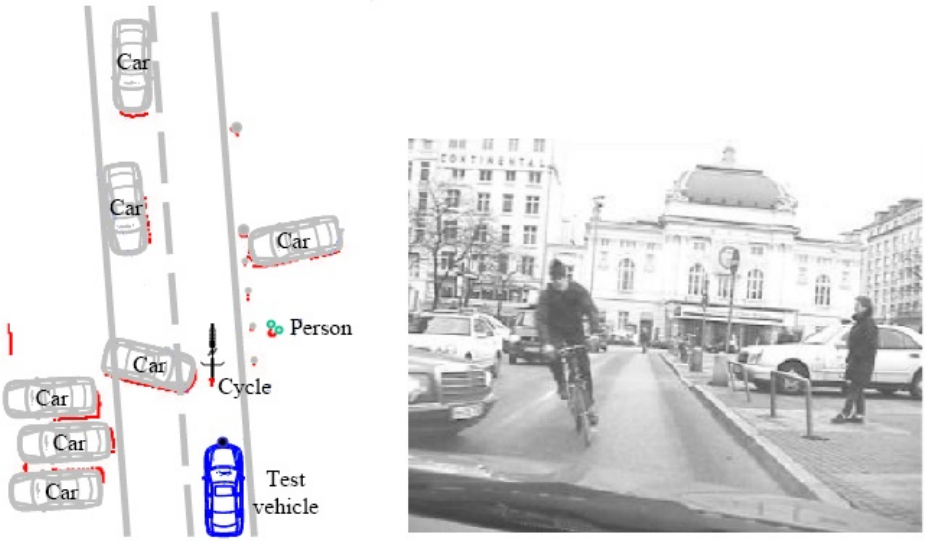
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Outline

1. Sensors and actuators
2. ROS
3. Perception
4. Decision
5. Action
6. Examples of applications
7. Conclusion

Sensor (1/3)

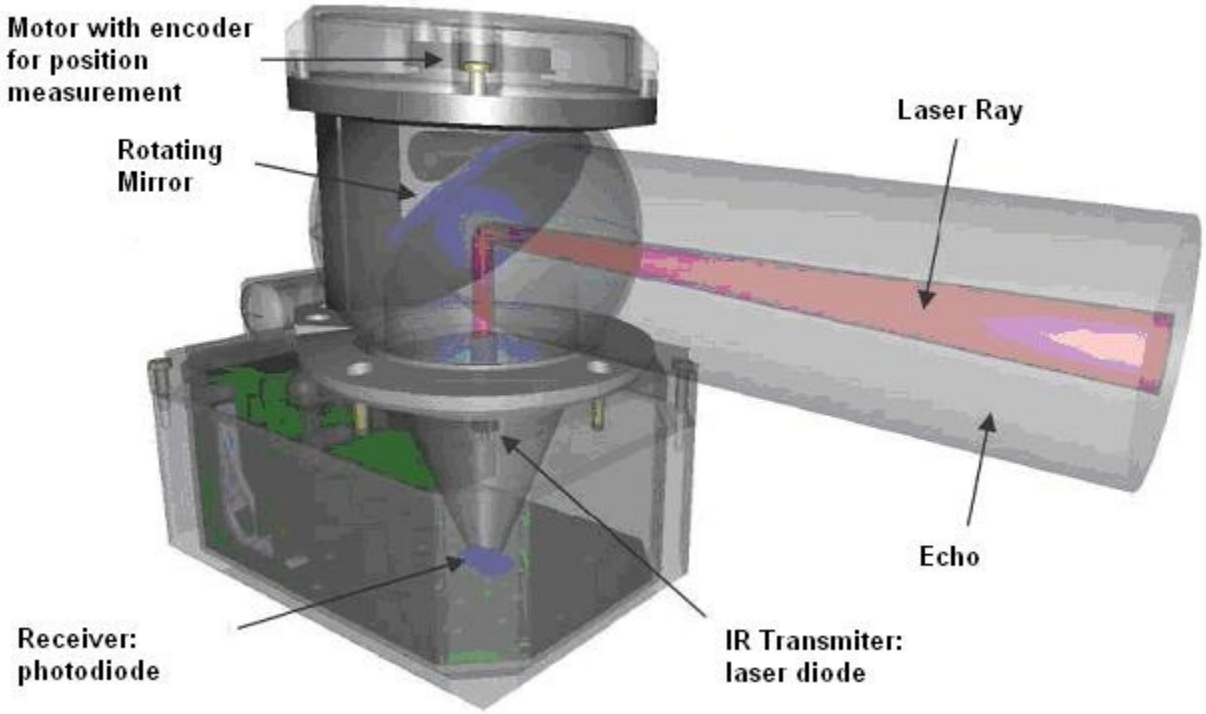
- **A sensor** is an instrument measuring a physical property of the environment;
- Sensors are imprecise and limited;
- The environment of a robot is generally complex, changing, unpredictable and uncertain;
- Understanding the world in which a robot evolves remains a challenge.



Courtesy of sick

Sensor (1/3)

- **A sensor** is an instrument measuring a physical property of the environment;
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Courtesy of sick

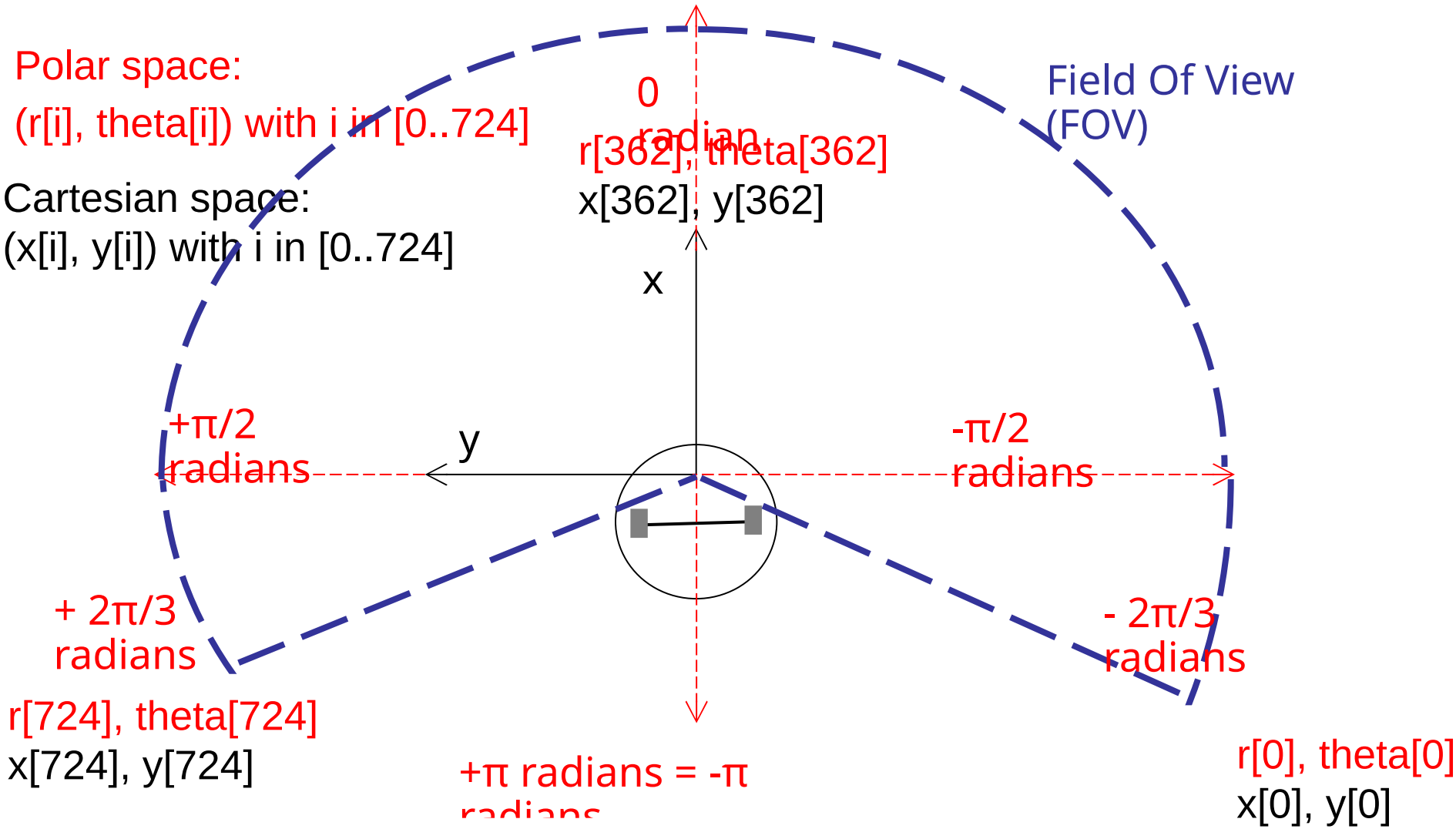
Sensor (2/3)

- **Robair** is equipped with a 2D laser scanner;
- The laser scanner has:
  - a range of about 5.5 meters;
  - a field of view of 240 degrees;
  - an angular resolution of 1/3 degrees.
- **Output**: a table with 725 elements (r, Θ)
- Laser data are acquired in the trigonometric way
- Polar to cartesian:
  - X = r cos (Θ);
  - Y = r sin (Θ).
- Quality of data depends on distance, angle…
- The frequence of acquistion is of about 40 Hz
- The laser scanner costs about 900 euros.



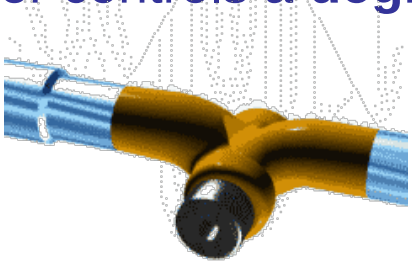
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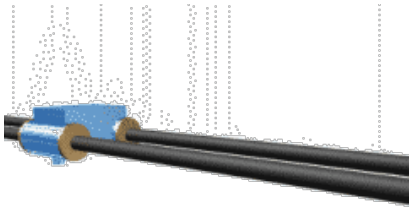
Sensor (3/3)


Polar space:
(r[i], theta[i]) with i in [0..724]
Cartesian space:
(x[i], y[i]) with i in [0..724]
0 radian
r[362], theta[362]
x[362], y[362]
Field Of View (FOV)
x
y
+π/2 radians
-π/2 radians
+ 2π/3 radians
- 2π/3 radians
r[724], theta[724]
x[724], y[724]
+π radians = -π radians
r[0], theta[0]
x[0], y[0]

Actuators of a mobile robot

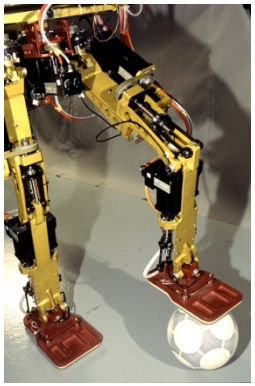
- An **actuator** is a component of a machine that is responsible for moving or controlling a mechanism or system;
- An actuator controls a **degree of freedom** (rotation, translation);





- Actuators could be complex**.**





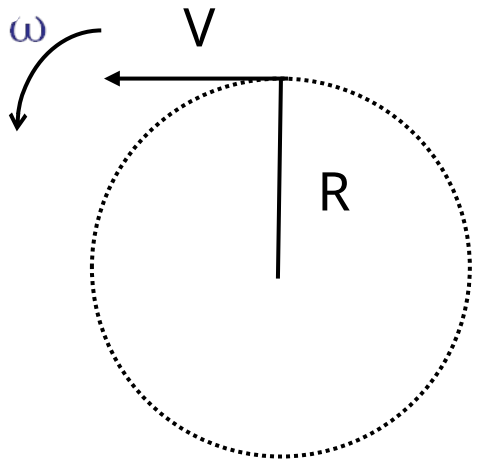


Courtesy of Thierry Fraichard

Actuators of robair (1/3)

- **Robair** has 2 wheels controlled by 2 motors;
- Robair is a differential drive robot;
- Simplest and most used kinematic model of robot.



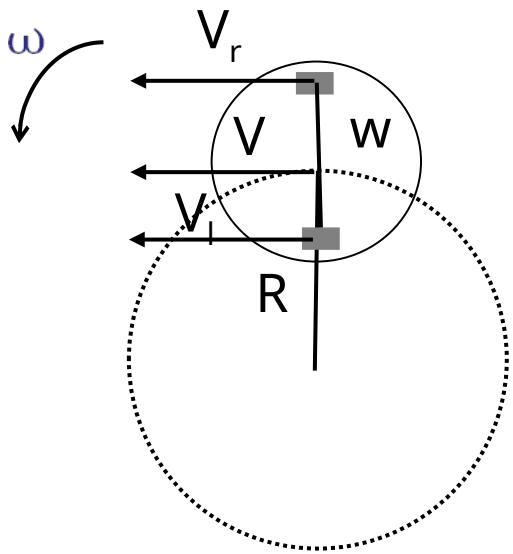


- **A monocycle** describes a virtual circle of radius R;
- We have $V = R\,\omega$

Actuators of robair (2/3)

- **Robair** has 2 wheels controlled by 2 motors;





- We have $V = R\,\omega$
- We have $V_r = (R + w/2)\,\omega$
- We have $V_l = (R - w/2)\,\omega$

- We find:

$$R = \frac{W}{2}\frac{V_l + V_r}{V_r - V_l}$$

Actuators of robair (3/3)

- Finally, we have the direct kinematic model:

$$\omega = \frac{V_r - V_l}{w} \quad (1) \qquad V = \frac{V_l + V_r}{2} \quad (2)$$

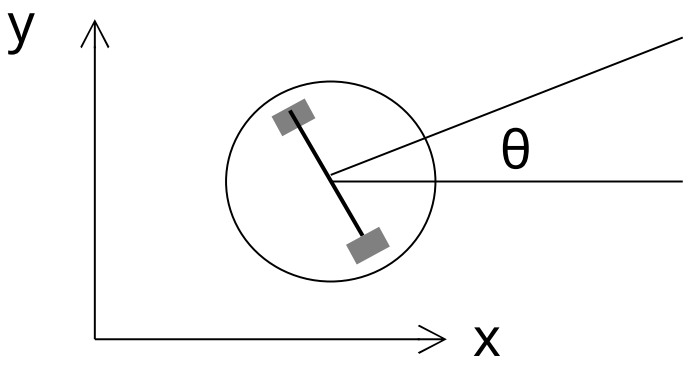
- Controlling (V_l, V_r), we can determine (V, ω)

$$V_r = V + \frac{w}{2}\omega \qquad V_l = V - \frac{w}{2}\omega$$

- To simplify the control (at the beginning), we will perform translation OR rotation in place

Estimation of motion: encoder

- While Robair is moving in its environment, we would like to know its position in this environment;
- Its position is determined by its position (x, y) in the environment + its orientation θ: (x, y, θ)
- X-axis is aligned with the angle 0 of the laser



- On each wheel, there is a system (named encoder) able to estimate the distance traveled by each wheel over a short time Δt

Estimation of position: odometry (1/7)

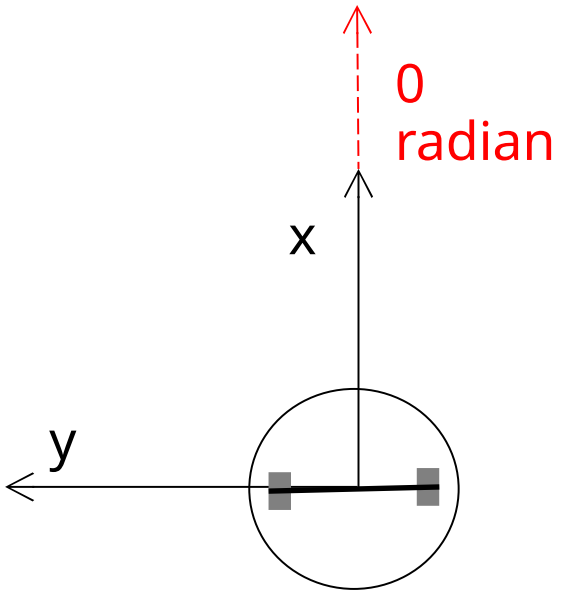
- We call d_l and d_r the distance traveled by each wheel over

$$d = \frac{d_r + d_l}{2} \text{ using (2)} \qquad \theta = \frac{d_r - d_l}{w} \text{ using (1)}$$

- $x_t = x_{t-1} + d\cos(\theta)$
- $y_t = y_{t-1} + d\sin(\theta)$
- $\theta_t = \theta_{t-1} + \theta$

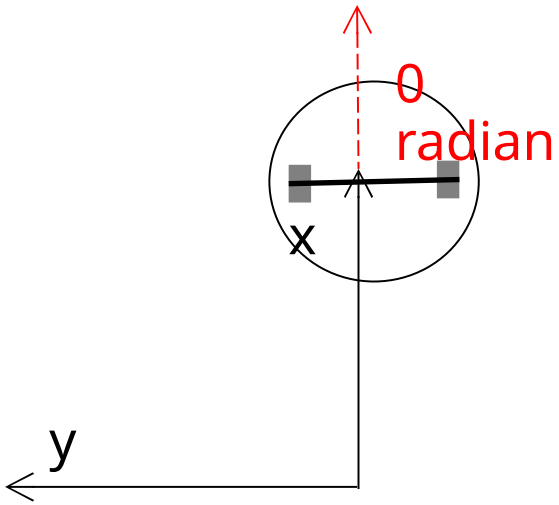
Estimation of position: odometry (2/7)

- When we start robair, odometry is (x=0m, y=0m,



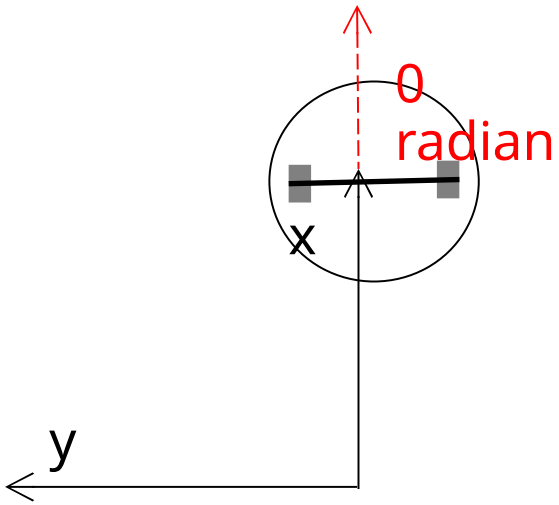
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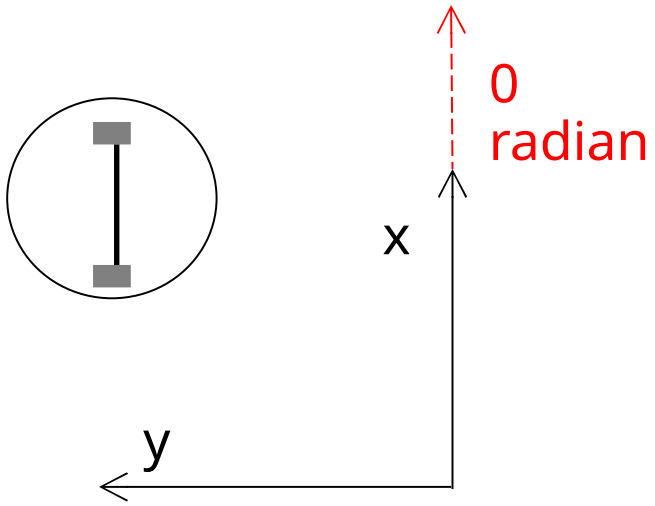
Estimation of position: odometry (4/7)

- When we start robair, odometry is (x=0m, y=0m,



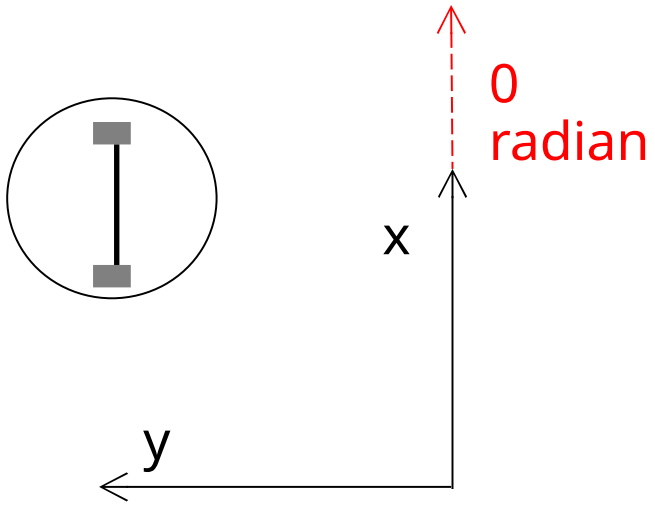
Estimation of position: odometry (5/7)

- When we start robair, odometry is (x=0m, y=0m,



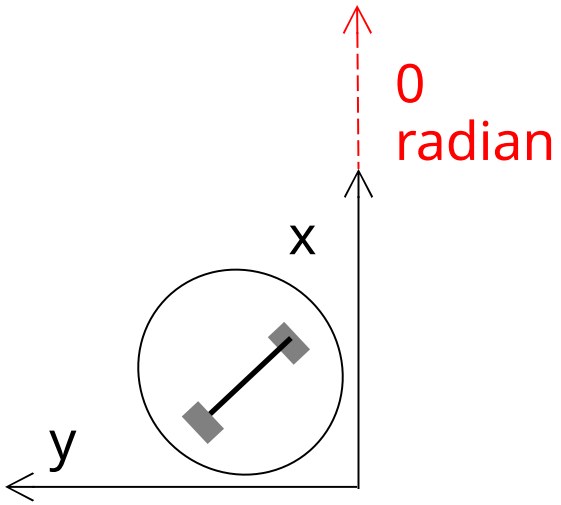
Estimation of position: odometry (6/7)

- When we start robair, odometry is (x=0m, y=0m,



Estimation of position: odometry (7/7)

- When we start robair, odometry is (x=0m, y=0m,



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ROS in a nutshell

- **ROS (Robot Operating System)** is a middleware for controlling robotic components from a PC: robots.ros.org;
- ROS is open source and a standard for software architecture development in robotics;

- ROS is based on 2 important concepts:
  1. A number of independent nodes;
  2. Messages (or topics) that are published by some nodes and subscribed by some nodes;
     => Messages are used to exchange information between nodes;

ROS in a nutshell

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- Next slides introducing ROS are based on 89-685: Introduction to Robotics (biu.ac.il) (lecture 1, slide 8-11)

Robots using ROS

- http://wiki.ros.org/Robots


Fraunhofer IPA Care-O-bot


Videre Erratic


TurtleBot


Aldebaran Nao


Lego NXT


Shadow Hand


Willow Garage PR2


iRobot Roomba



Robotnik Guardian


Merlin miabotPro


AscTec Quadrotor


CoroWare Corobot



Clearpath Robotics Husky



Clearpath Robotics Kingfisher


Festo Didactic Robotino

ROS nodes

- Single-purposed executable programs
  - e.g. sensor driver(s), actuator driver(s), mapper, planner, UI, etc.
- Individually compiled, executed, and managed
- Nodes are written using a ROS **client library**
  - C++ client library
  - python client library (not provided in this course)
- Nodes can publish or subscribe to a Topic

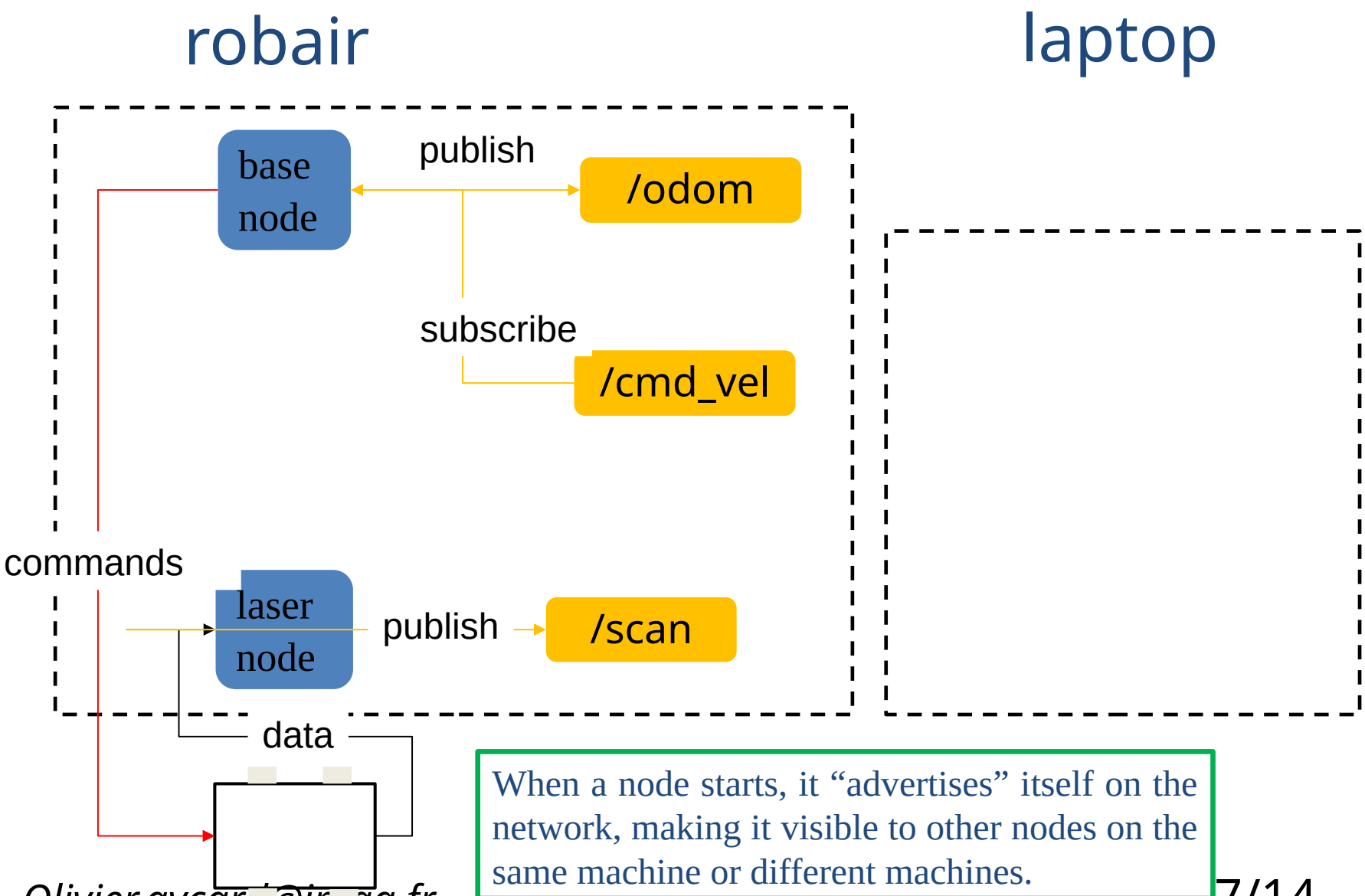
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ROS topics

- A topic is a name for a stream of messages with a defined type
  - e.g., data from a laser range-finder might be sent on a topic called scan, with a message type of LaserScan
- Nodes communicate with each other by publishing messages to topics
- Publish/Subscribe model

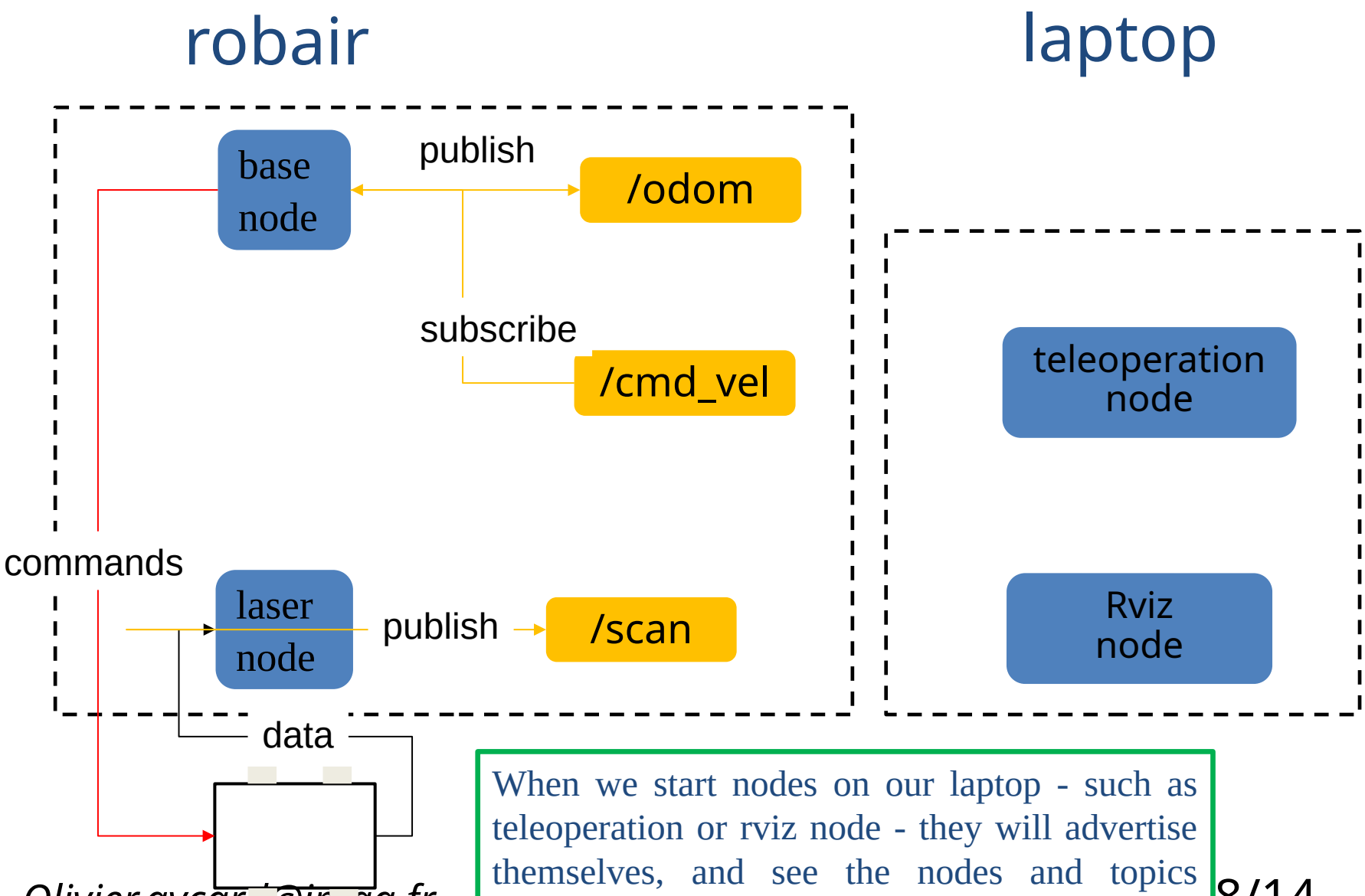
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ROS on robair (2/5)



When a node starts, it “advertises” itself on the network, making it visible to other nodes on the same machine or different machines.

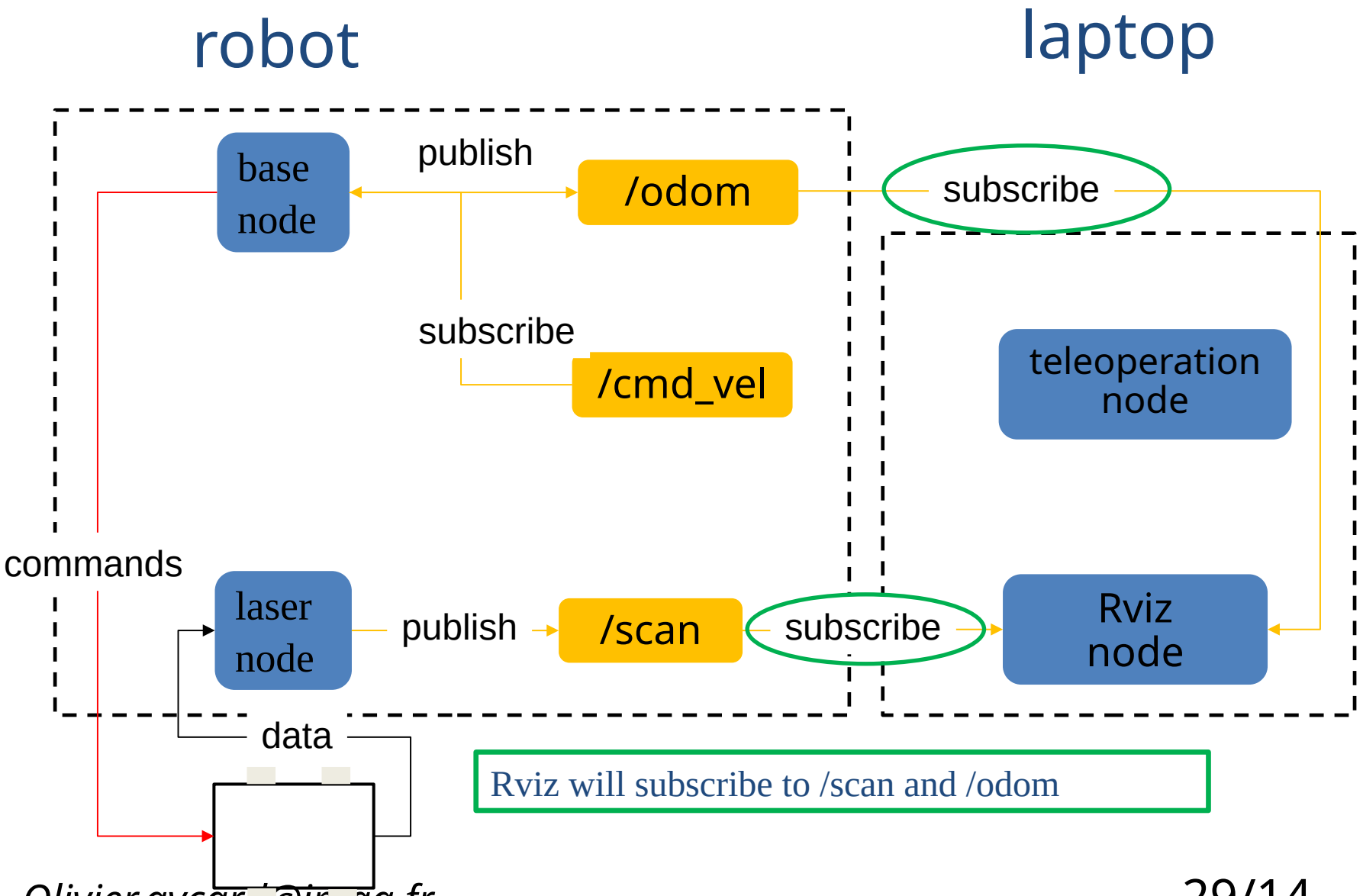
ROS on robair (3/5)



When we start nodes on our laptop - such as teleoperation or rviz node - they will advertise themselves, and see the nodes and topics running on RobAIR.

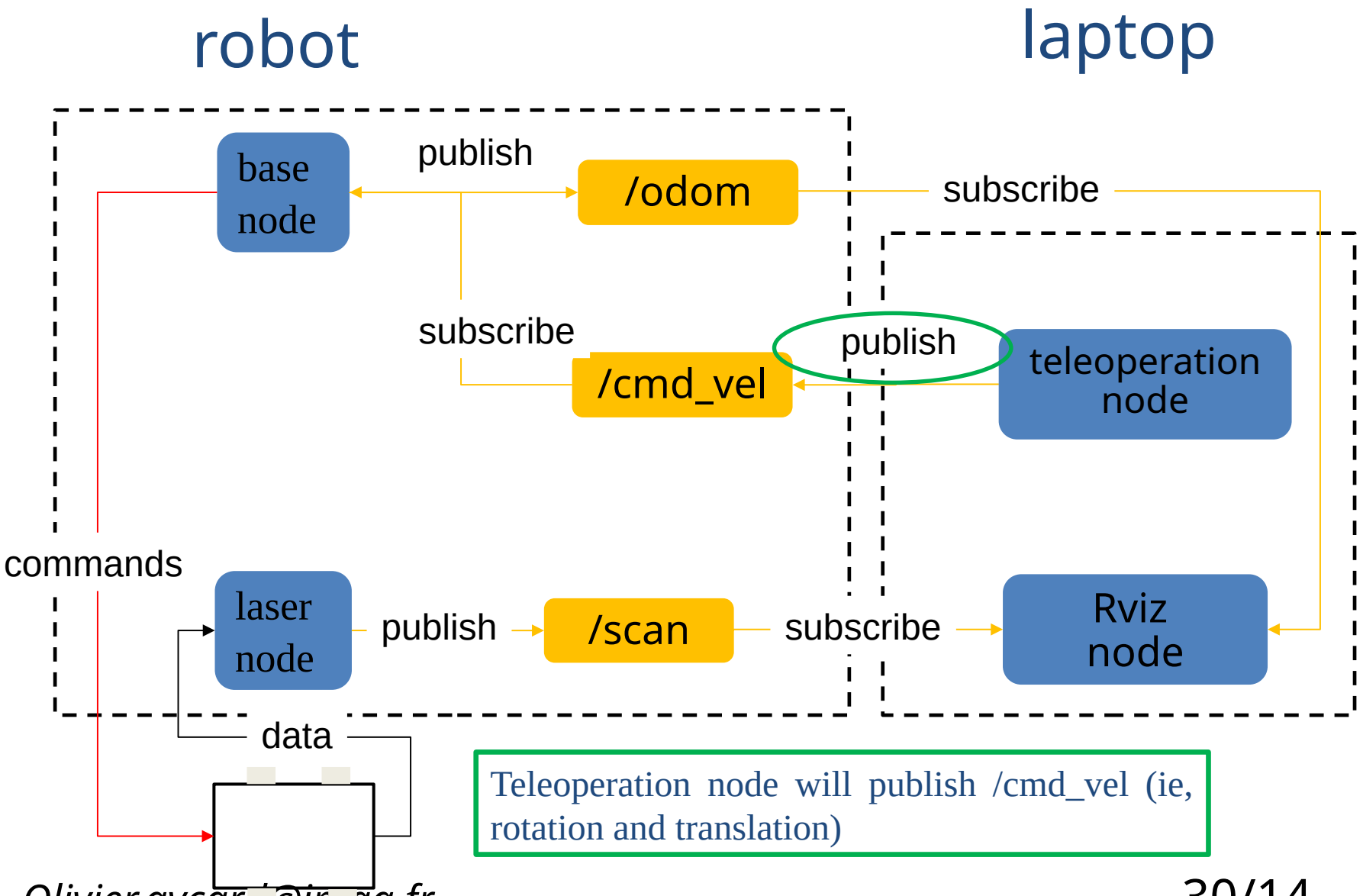
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ROS on robair (4/5)



Rviz will subscribe to /scan and /odom

ROS on robair (5/5)



Teleoperation node will publish /cmd_vel (ie, rotation and translation)

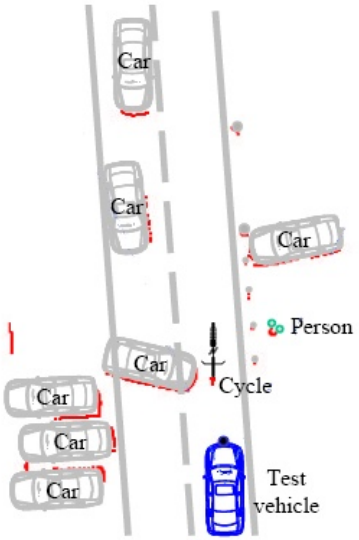
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Perception

Goal

- Robot perception in dynami environments
- **Laser scanner**
- Speed and robustness





Present Focus: interpretation of raw and noisy sensor data

- Identify static and dynamic part of sensor data
- Modeling dynamic part of the environment
  - Detection And Tracking of Moving Objects (DATMO)
- Modeling static part of the environment
  - Simultaneous Localization And Mapping (SLAM)

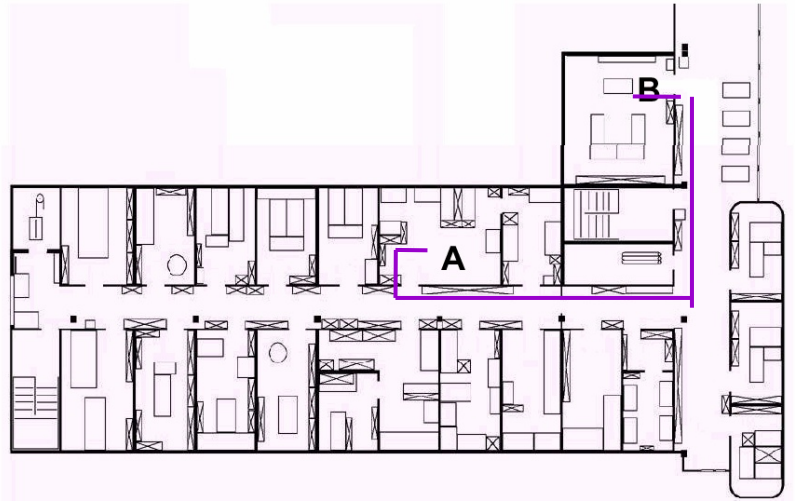
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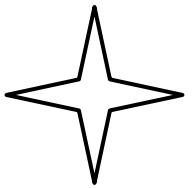
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Decision/Plan of future actions

- Most of the time, a mobile robot has to move in its environment:
  - It needs to plan its future actions
- The mobile robot has a map and it knows where it is in the map (position A for instance);
- It should reach an other position in the map (position B for instance)
- **Question**: How to get there ?
- Answer: sequence of actions to go from A to B that is feasible and without collisic





West, South, East (12 times), North (6 times), West (twice)

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Action/control/navigation

- The mobile robot has a sequence of actions to execute, to reach its goal: it has to execute this sequence of actions:
  - Typical action in our case: “move to (x, y)”;
  - Monitoring of execution: we monitor what happen and react if needed.
    - We need to be able to estimate actions/motions of the mobile robot;
  - Collision detection/avoidance: the mobile robot should be able to detect (and avoid) collision.

Examples of applications(1/3)

- Advanced driver assistant system (ADAS) or autonomous vehicles



Darpa Urban Challenge 2007



Google car



IP Prevent 2008



Google car 2016

Examples of applications(2/3)

- Service robotics



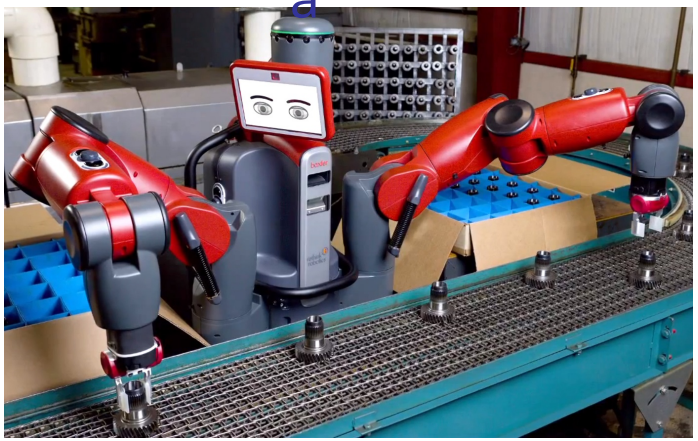
Roomba



Robomow



Dyia One



Baxter



Staubli

Examples of applications(3/3)

- Companion robots



Paro



Aibo



Buddy



Nao



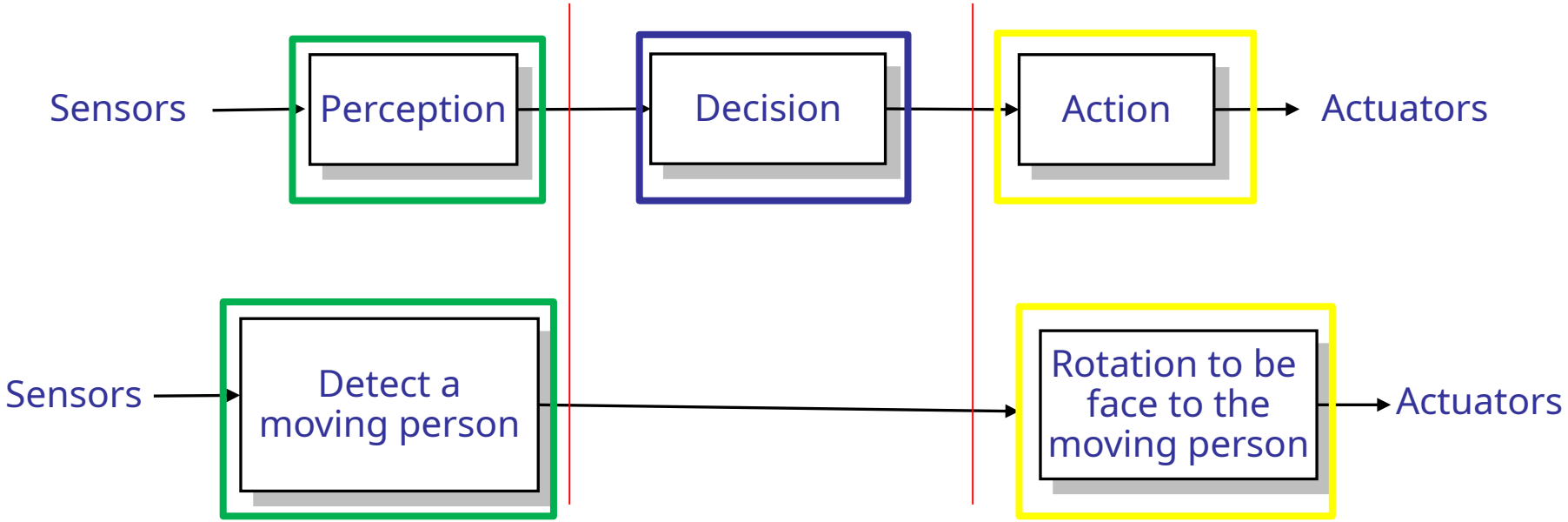
Pepper

Conclusion

- A mobile robot is equipped with 2 kind of sensors:
  - **Exteroceptive sensors** that give information about the environment (ie, laser scanner);
  - **Proprioceptive sensors** that give information about the internal state of the robot (ie, odometer);
- A mobile robot is equipped with some actuators characterized by their degree of freedom;
  - Robair is a differential drive robot;
- Sensors and actuators are imprecise and limited;
- The environment of a robot is generally complex, changing, unpredictable and uncertain.

Follow me behavior (1/2)

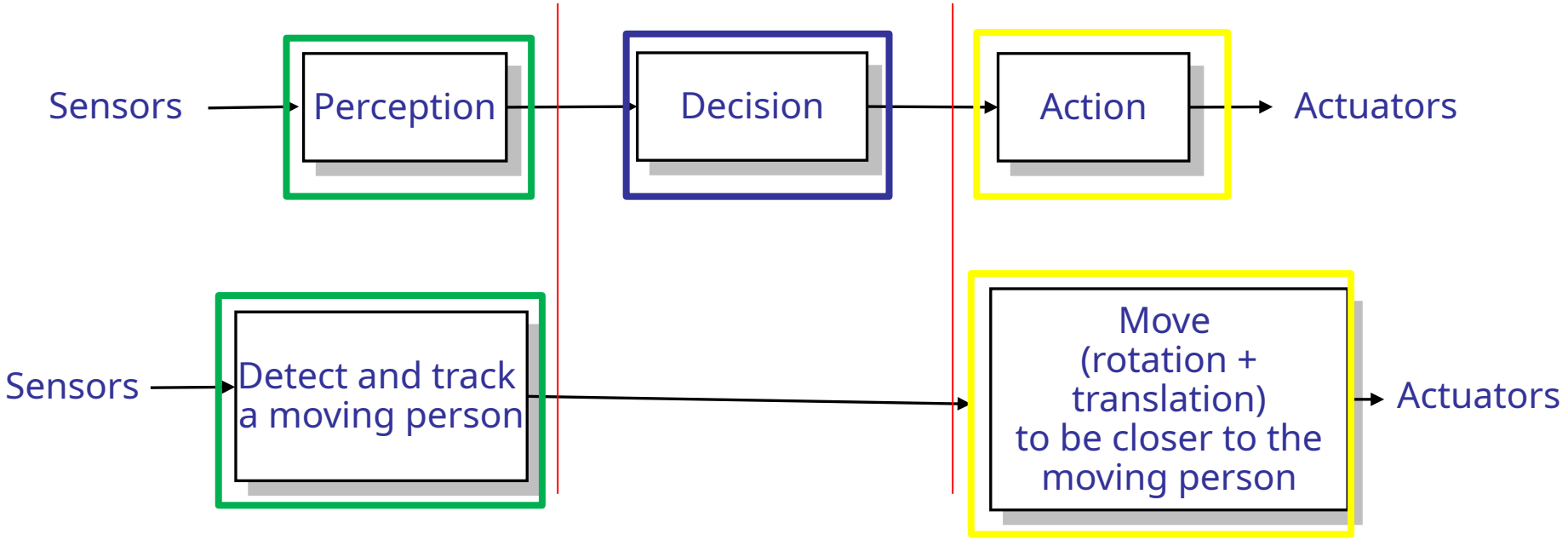
- We design, implement and test a simple “follow me” behavior in the next weeks in 2 steps



- 1st release of “follow me” behavior

Follow me behavior (2/2)

- We design, implement and test a simple “follow me” behavior in the next weeks in 2 steps



- 2nd release of “follow me” behavior